

PatientTriage.ai

“Triage is a snapshot. Risk isn't.”

AI recommends. Deterministic safety rules protect. Clinicians decide.

AI Safety Copilot for the Emergency Waiting Room • Active Clinical Decision Support • Accenture Challenge 2026

COMPETITION TRACK Round 2 Technical Prototype	CORE STACK Python 3.13 / FastAPI / React 19	VERIFICATION 51/51 Tests Passed (100%)	REPOSITORY github.com/freya1705/PatientTriageAI
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1. The Post-Triage Clinical Surveillance Gap

Triage determines priority at the front door—but patients don't remain static while they wait. ESI Level 3/4 patients wait 2.5 to 4.5 hours unmonitored before physician examination, during which silent decompensation (hypoxia, sepsis, internal hemorrhage) frequently remains undetected. **PatientTriage.ai** is an **AI Safety Copilot** that closes this post-triage gap by continuously identifying deterioration, stale evidence, and uncertainty, then prioritizing who clinicians should reassess first.

2. Continuous Patient Journey: Dynamic Priority Lifecycle

Patient Arrives ■■■ Intake Triage ■■■ Safe-to-Wait Baseline ■■■ Waiting Lounge (2.5–4.5h)

Bedside Reassess ■■■ Ranked Action Queue ■■■ Re-Evaluate (■/■/■/■) ■■■ Trajectory & Staleness Drift
Key Operational Innovation: The patient's clinical priority updates dynamically after intake triage based on real-time safety trajectory.

3. Three Core Architectural Differentiators

- 1. Safe-to-Wait Dynamic Surveillance:** Continuously tracks physiological trajectory velocity (ΔSpO_2 , ΔHR) over time rather than freezing static intake numbers.
- 2. Uncertainty Guardrail (*Unknown* $\neq$ *Safe*):** When critical observations are missing or stale, conventional workflows may lack sufficient evidence to safely reassess waiting patients. PatientTriage.ai applies an explicit uncertainty penalty ($w_u = +15$ to $+25$ pts), elevating unmonitored patients for human physical verification.
- 3. Attention Gap Optimization:** Traditional queues sort strictly by intake severity. PatientTriage.ai discounts attended cases ($w_c = -35$ pts) to elevate unmonitored deteriorating waiting patients directly to Rank #1.

4. Four Discrete Operational Workflow States

Badge & State	Physiological Definition	Clinical Action & System Dispatch
■ CONTINUE	Vitals stable within baseline limits; observation shelf-life active.	Safe to continue waiting; ongoing ambient telemetry surveillance.
■ REASSESS	Observation shelf-life expired (stale data) or moderate drift ($\Delta HR \geq +20$ bpm).	Dispatches nurse bedside check to refresh vitals and re-evaluate trajectory.
■ ESCALATE	Deterministic red-flag breach ($SpO_2 < 85\%$, $SBP < 75$ mmHg) or rapid collapse.	Immediate resuscitation bay allocation and physician/trauma notification.
■ UNCERTAIN	Incomplete vital telemetry, sensor disconnect, or missing intake parameters.	Forces human vital capture under <i>Unknown</i> $\neq$ <i>Safe</i> to prevent under-triage.

5. System Architecture: Core Engine vs. Phase 4 Extensions

Tier	Core Waiting-Room Safety Scope	Phase 4 Future Extensions
Tier 1: Guardrails	Deterministic red-flags ($SpO_2 < 85\%$, $SBP < 75$, stroke) and downgrade safety blocking. Bypasses statistical layer.	Regional emergency safety network federation.
Tier 2: Intelligence	Vital velocity calculation, confidence decay, uncertainty penalty calibration, and Attention Gap prioritization formula.	108 EMS Pre-Arrival FHIR telemetry & Referral Candidate Scoring (RES).
Tier 3: Dispatch	Ranked Action Queue ('Next 5 Mins'), 1-click Bedside Reassessment, and tamper-evident append-only audit trail.	Regional community clinic referral diversion & multi-facility network dashboarding.

6. Attention Gap Formula & Plain-English Operational Logic

Action Priority Score = (w_r × Risk) + (w_d × Deterioration) + (w_s × Staleness) + Wait Hazard + (w_u × Uncertainty) – (w_c × Clinical Coverage)

Operational Translation: The system does not ask only 'Who is most sick?' It asks 'Who needs clinical attention most urgently right now, considering trajectory collapse, stale data, uncertainty, and whether a clinician is already at the bedside?'

Calibrations: w_r = 1.0 (base risk) • w_d = +25 to +40 pts (ΔSpO₂ ≤ -5%, ΔHR ≥ +20 bpm) • w_s = +20 to +35 pts (safety clock expiry) • w_u = +15 to +25 pts (missing vitals) • w_c = -35 pts (attended discounting).

7. Competitive Moat vs. Conventional Clinical Workflows (Epic / Cerner)

Evaluation Dimension	Conventional Clinical Workflows	PatientTriage.ai AI Safety Copilot
Surveillance Domain	Admitted inpatient beds; static snapshot at intake.	Dedicated waiting-room surveillance layer.
Missing Vitals Handling	Lacks sufficient evidence to safely reassess.	<i>Unknown ≠ Safe</i> : increases uncertainty penalty.
Clinical Coverage Factor	Ignores whether patient is attended or waiting alone.	Attention Gap discounts attended cases to surface waiting needs.
Deterioration Detection	Threshold alarms only after severe boundary breach.	Tracks trajectory velocity (ΔVitals/Δt) before collapse.

8. Prototype Benchmark Evaluation (20 Synthetic Test Cohorts & 51 CI/CD Tests)

Disclaimer: These prototype benchmark results validate algorithmic behavior and guardrail execution only; they are not evidence of clinical efficacy or diagnostic accuracy.

Safety Dimension	Static Intake Triage	PatientTriage.ai Prototype	Algorithmic Impact
Waiting Deterioration Catch Rate	0 / 20 detected	20 / 20 detected (100%)	Surfaces hidden decompensation
Stale Observation Flagging	0 / 20 flagged	20 / 20 flagged (EXPIRED)	Zero unmonitored stale waits
Missing Vitals Under-Triage	High (lacks evidence)	0% False Reassurance	Forces physical human recheck
Attention Gap Priority Re-Rank	None (attended block)	Active Queue Re-Ranking	Optimizes scarce nurse attention
Unsafe Downgrade Prevention	0 Guardrails	100% Guarded (Blocked)	Deterministic safety floor

9. Phased Clinical Implementation Roadmap & Trial Endpoints

- **Phase 1 (Q3 2026 - Completed):** Lab Benchmark Validation • 20 synthetic cohorts verified across 51 automated pytest test cases; sub-15ms inference latency.
- **Phase 2 (Q4 2026):** Shadow Clinical Trial • Non-interventional background FHIR integration alongside Epic/Cerner to evaluate clinician concordance.
- **Phase 3 (Q1–Q2 2027):** Live Pilot • Target Endpoints (Requiring Clinical Validation): >45% reduction in Mean Time to Escalation (MTTE); false alarm rate < 2 alerts/nurse/shift; ≥25% reduction in Left-Without-Being-Seen (LWBS).
- **Phase 4 (Q3 2027+):** Multi-Hospital Enterprise Scope • Regional load balancing, 108 EMS ambulance telemetry ingestion, and community referral diversion.

10. Quick Start & Regulatory / Fail-Safe Architecture

Execution: .\start.ps1 | Test Suite: python -m pytest -v (51 Passed) | UI Cockpit: http://localhost:5173

Regulatory Classification: PatientTriage.ai is an active clinical decision-support research prototype developed for the Accenture Innovation Challenge 2026. Positioned under FDA Non-Device CDS guidance (21 U.S.C. § 360aaa-1); licensed clinicians retain 100% decision authority. **Fail-Safe Protocol:** If network connection drops, the system defaults safe and prompts manual rounding with deterministic safety guardrails active.